Findings Ledger — Figure Variants (pick-and-choose)

FC–SC connectome translation · F1–F13 · both parcellations, all estimators

Adel Sahuc

How to use this deck

- One section per finding (F1–F13); each slide is one chart *variant* of that finding.
- Multiple chart types per finding (bars / hbar / lollipop / dumbbell / slope / heatmap / box, plus bespoke).
- Pick your favourites; each variant is also saved standalone in figures/slides/.

Provenance: F1–F5, F9–F13 from the frozen reproduction-grid CSVs + sanity outputs; F6–F8 from the separate family-mechanism pipeline (same frozen splits, adds the sibling/twin task).

Estimators: reconstruction uses all three; for *scalar* downstream targets the RBF KernelRidge rows are *a priori* ill-conditioned (1-D target, α/γ grid) and shown only to expose that instability — BayesianRidge is reportable.

The affirmative claim, stated at its defensible width

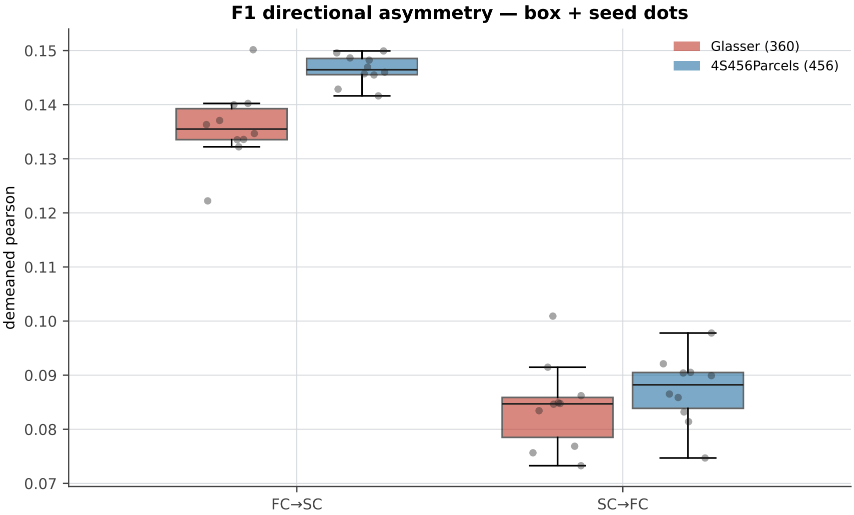
- The durable contributions are the **asymmetry (F1)** and the **baseline/ceiling negatives (F5/F9/F10/F13)**.
- The affirmative cognition claim is **narrow**: observed FC carries a **crystallized**-cognition signal that survives demographic residualization and restricted-family FDR in one cell (obs FC+bv+demo $\rightarrow$ CogCryst, $q = 0.027/0.048$).
- The broader "FC beats the baseline on cognition" lift is real but **multiplicity-fragile** (0/54 survive whole-grid FDR) — reported, not hidden (F11/F12).

F1 · Directional asymmetry (FC$\rightarrow$SC $>$ SC$\rightarrow$FC)

8 chart variants

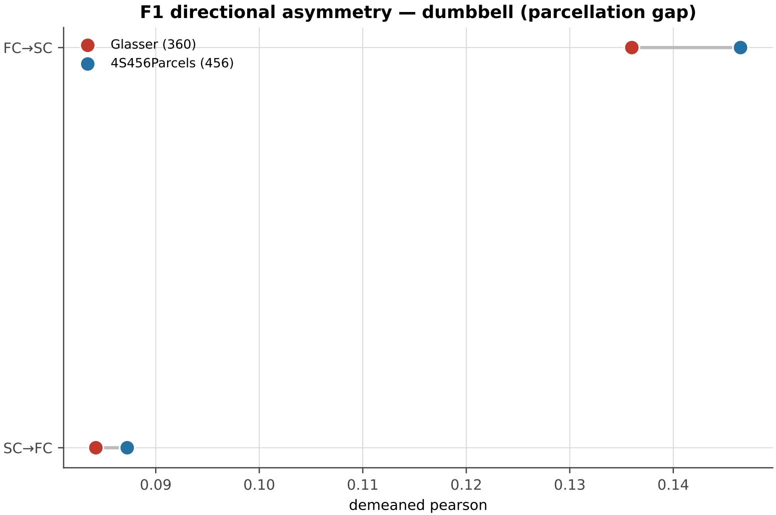
F1 · Directional asymmetry (FC→SC > SC→FC)

box + seed dots



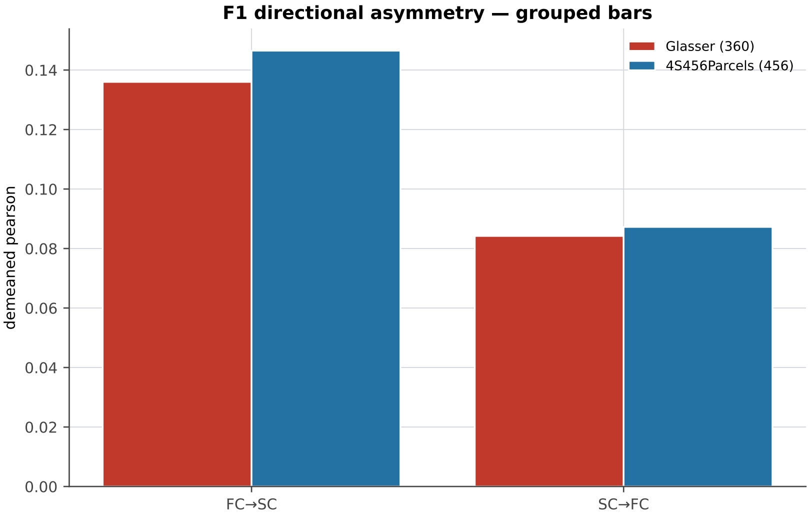
F1 · Directional asymmetry (FC→SC > SC→FC)

dumbbell (parcellation gap)



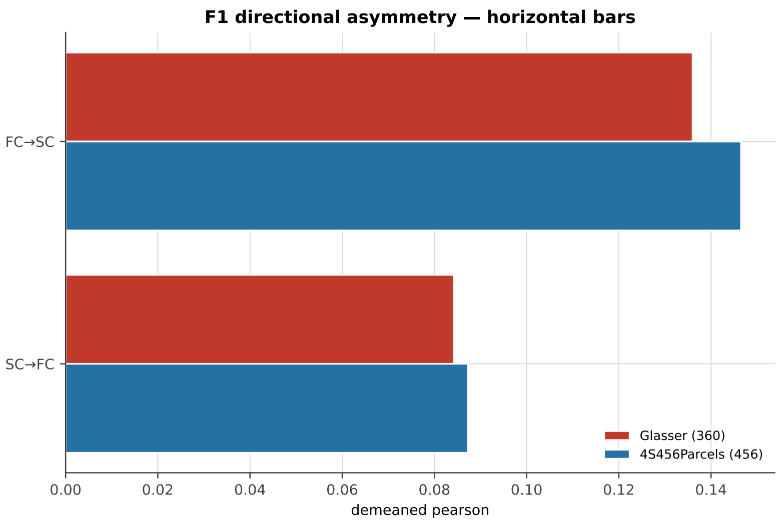
F1 · Directional asymmetry (FC→SC > SC→FC)

grouped bars



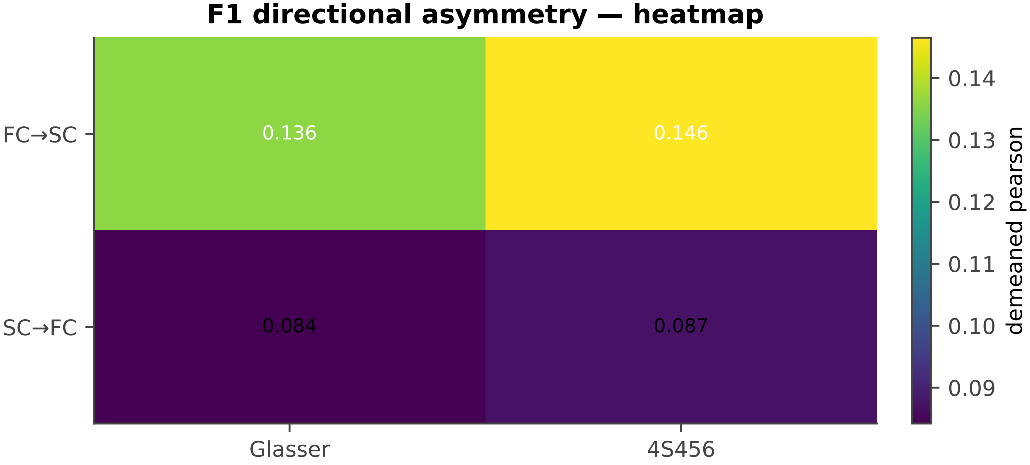
F1 · Directional asymmetry (FC→SC > SC→FC)

horizontal bars



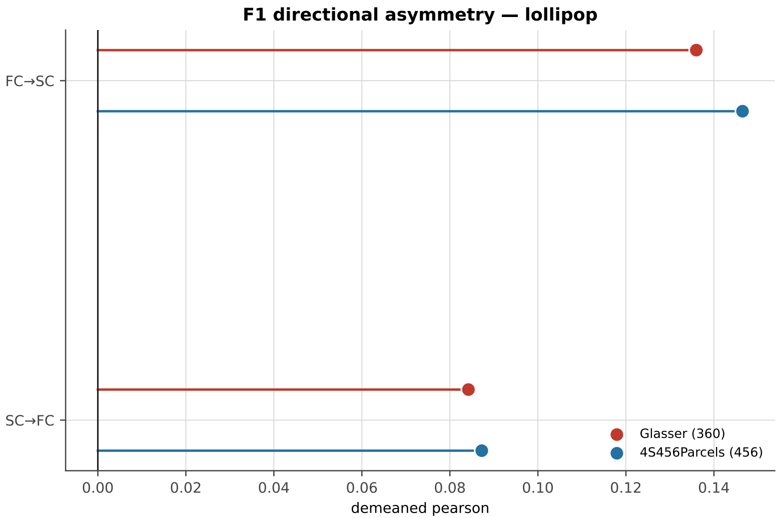
F1 · Directional asymmetry (FC→SC > SC→FC)

heatmap



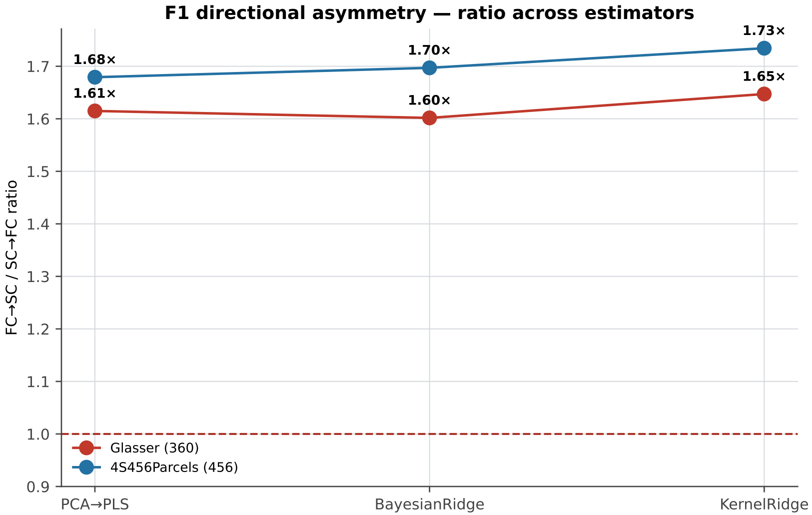
F1 · Directional asymmetry (FC→SC > SC→FC)

lollipop



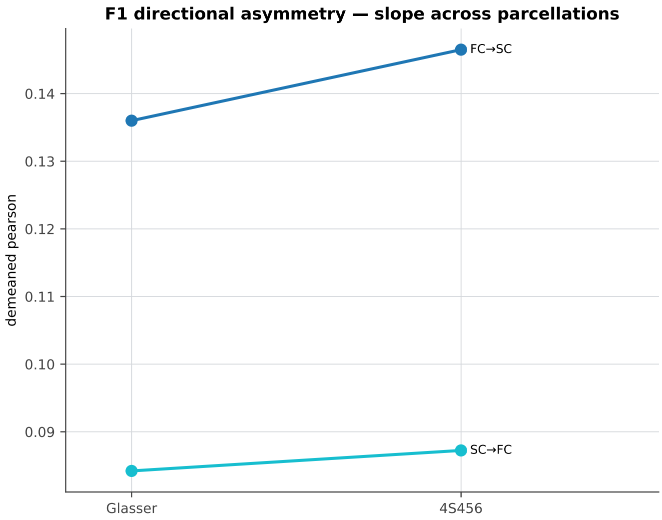
F1 · Directional asymmetry (FC→SC > SC→FC)

ratio across estimators



F1 · Directional asymmetry (FC→SC > SC→FC)

slope across parcellations

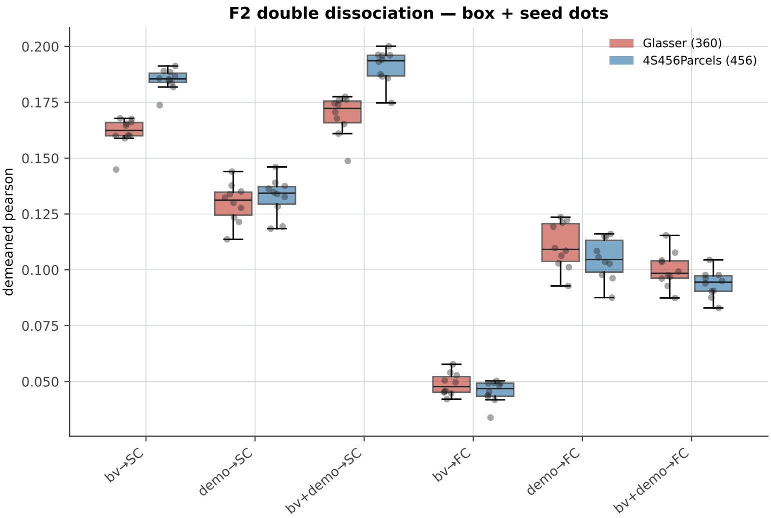


F2 · Anatomy$\rightarrow$SC / demographics$\rightarrow$FC dissociation

7 chart variants

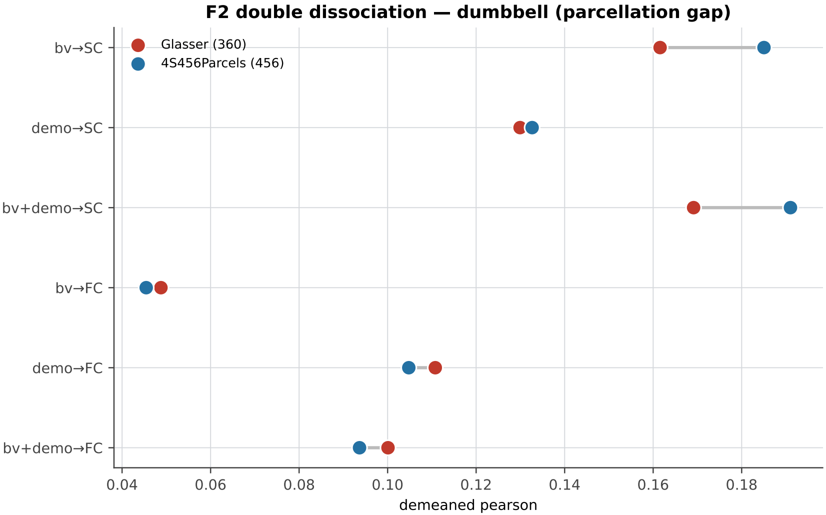
F2 · Anatomy→SC / demographics→FC dissociation

box + seed dots



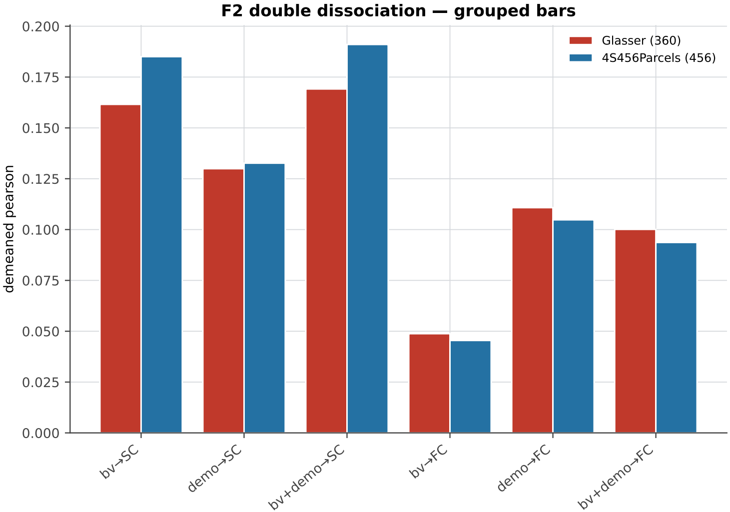
F2 · Anatomy→SC / demographics→FC dissociation

dumbbell (parcellation gap)



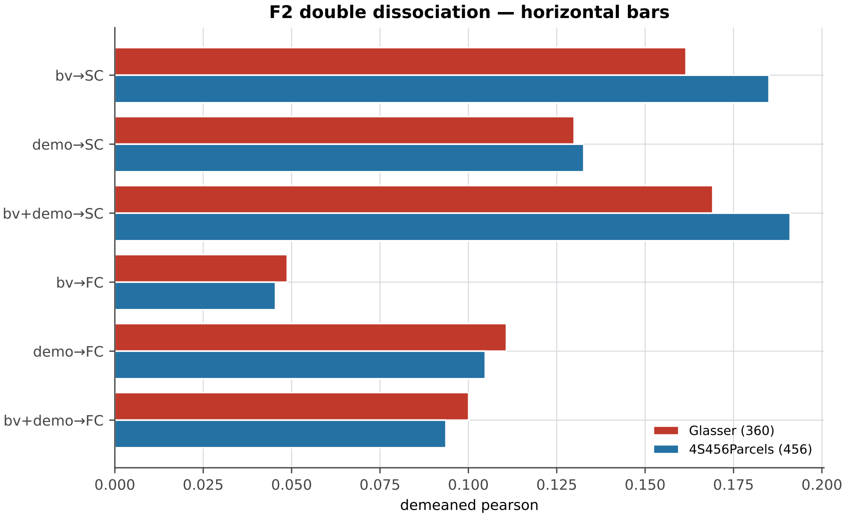
F2 · Anatomy→SC / demographics→FC dissociation

grouped bars



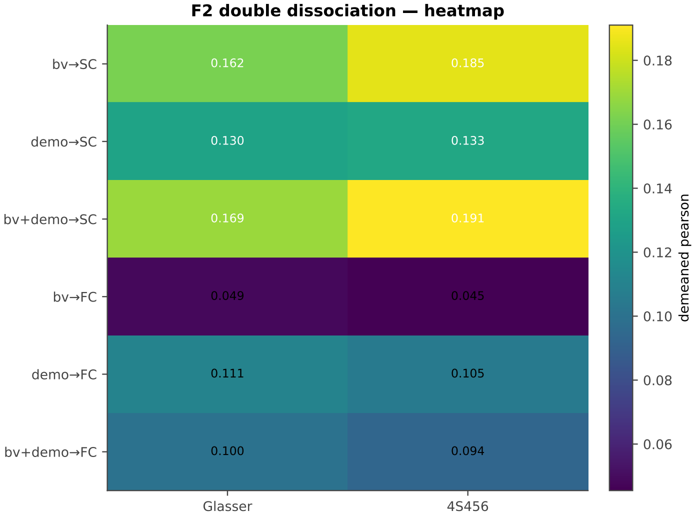
F2 · Anatomy→SC / demographics→FC dissociation

horizontal bars



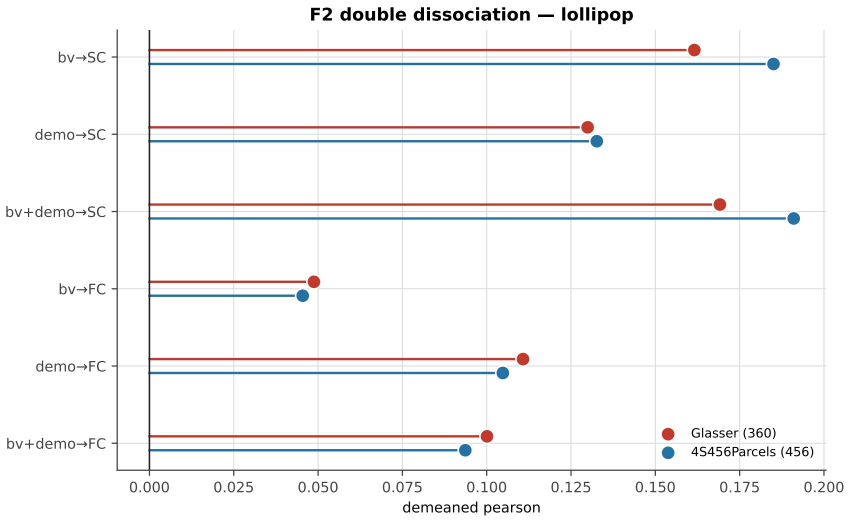
F2 · Anatomy→SC / demographics→FC dissociation

heatmap



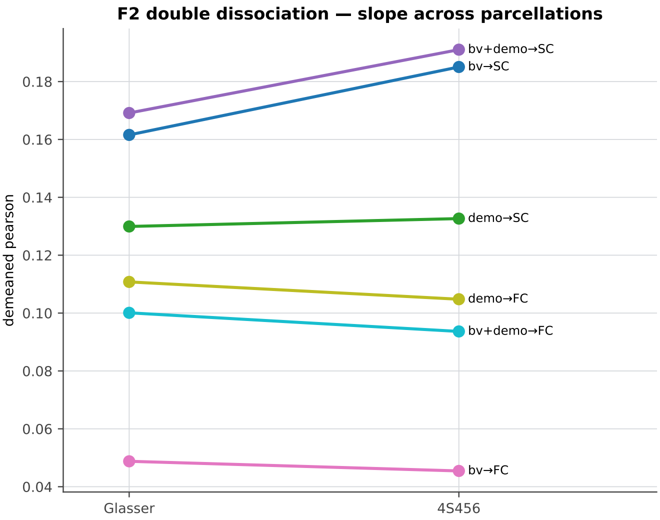
F2 · Anatomy→SC / demographics→FC dissociation

lollipop



F2 · Anatomy→SC / demographics→FC dissociation

slope across parcellations

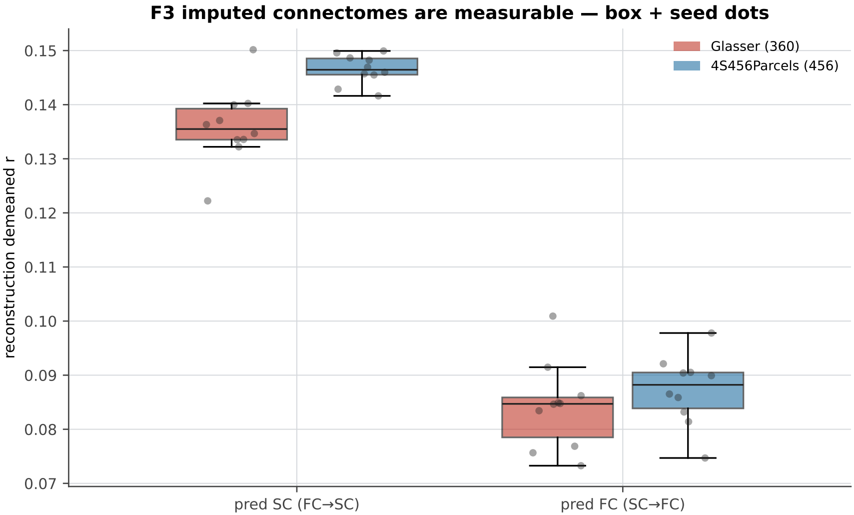


F3 · Imputed connectomes are measurable

7 chart variants

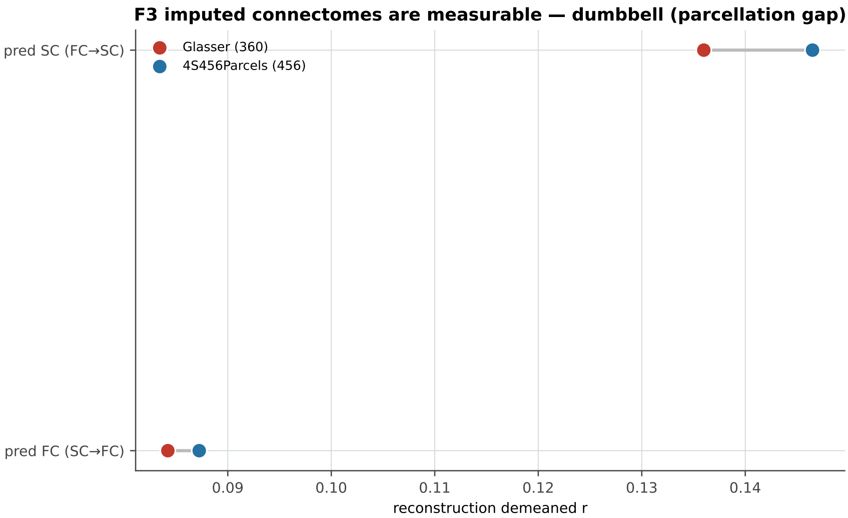
F3 · Imputed connectomes are measurable

box + seed dots



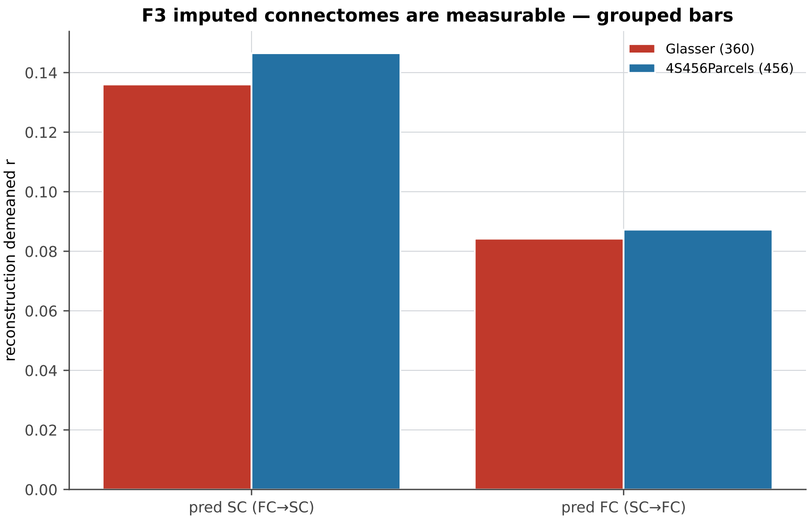
F3 · Imputed connectomes are measurable

dumbbell (parcellation gap)



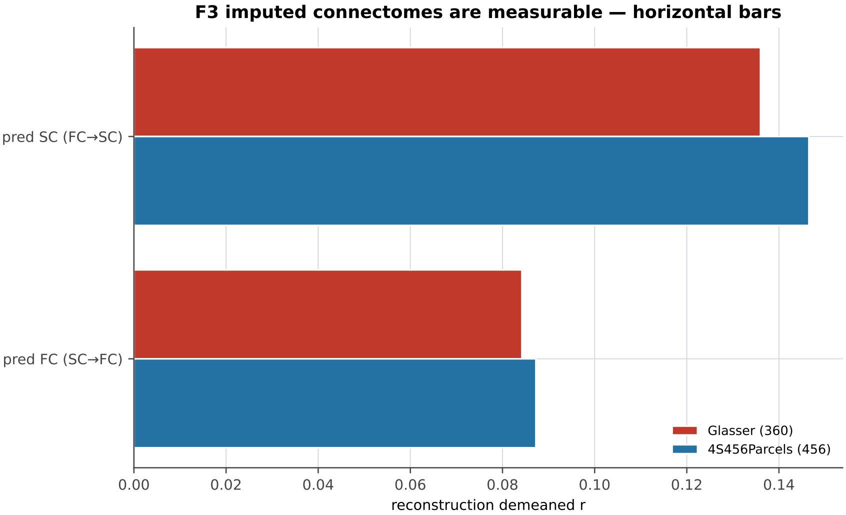
F3 · Imputed connectomes are measurable

grouped bars



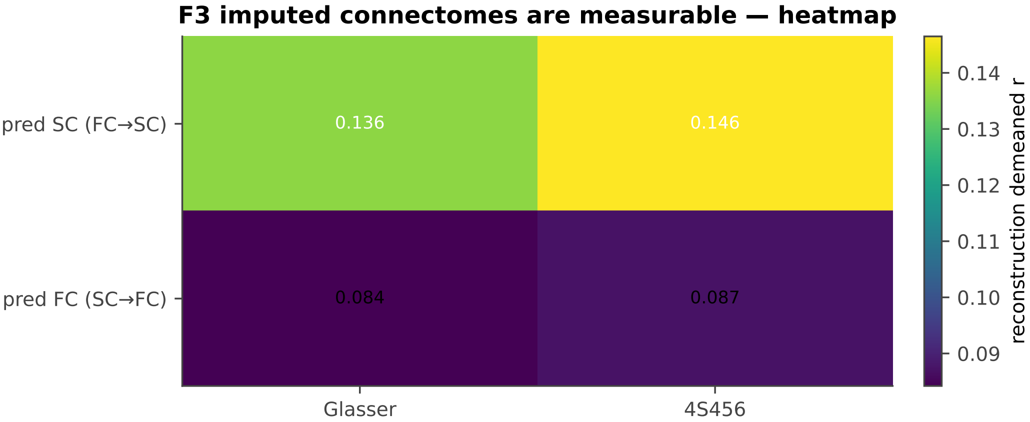
F3 · Imputed connectomes are measurable

horizontal bars



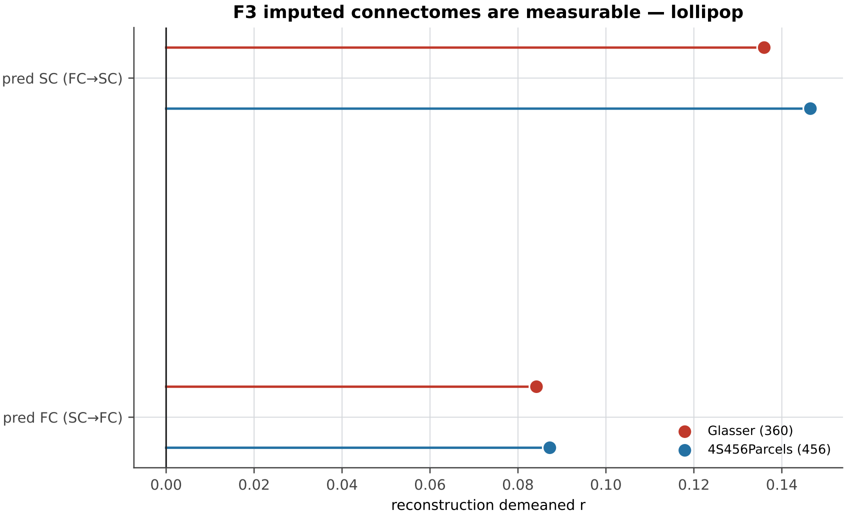
F3 · Imputed connectomes are measurable

heatmap



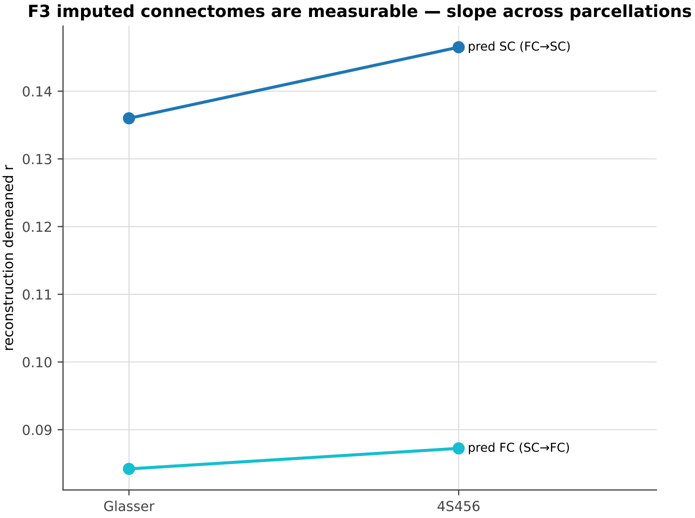
F3 · Imputed connectomes are measurable

lollipop



F3 · Imputed connectomes are measurable

slope across parcellations

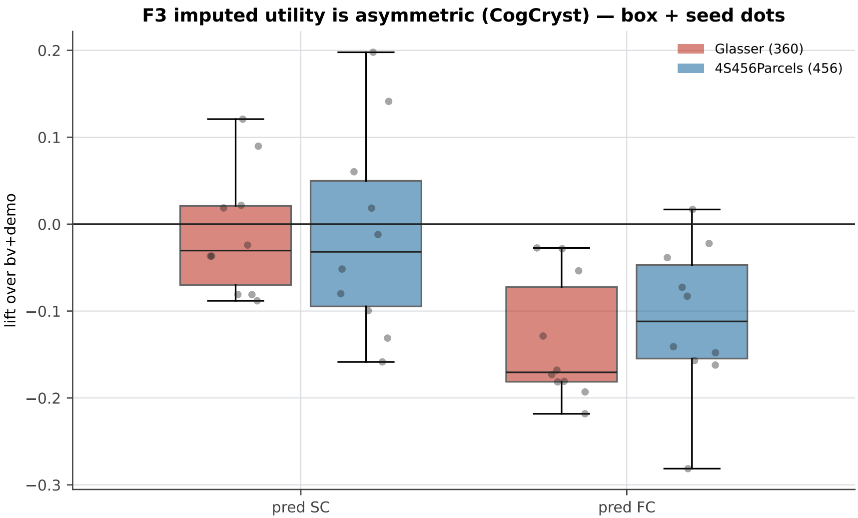


F3 · . . . but their utility is asymmetric

7 chart variants

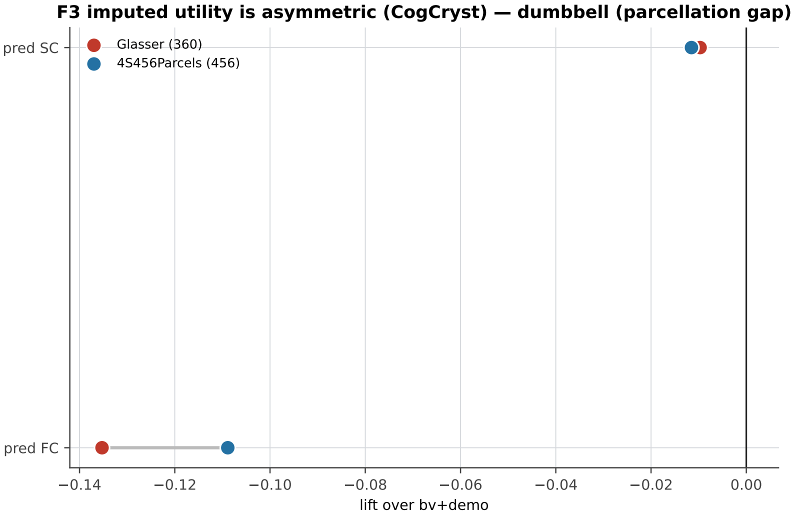
F3 · . . . but their utility is asymmetric

box + seed dots



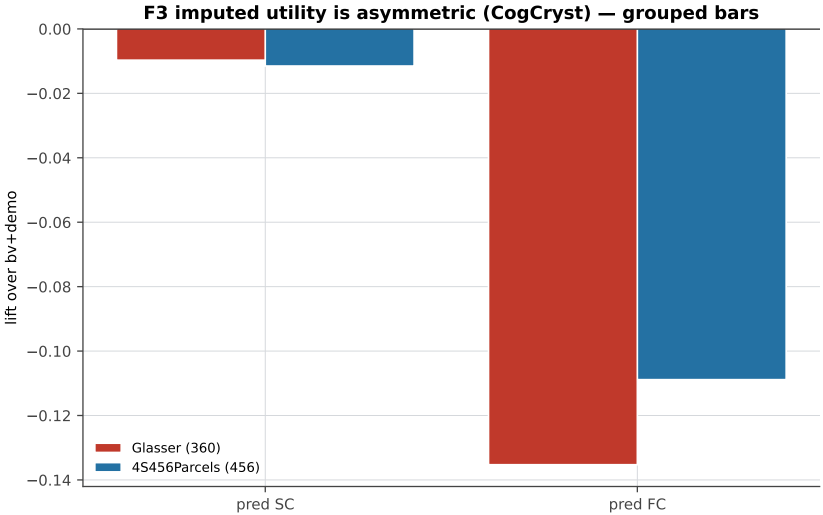
F3 · . . . but their utility is asymmetric

dumbbell (parcellation gap)



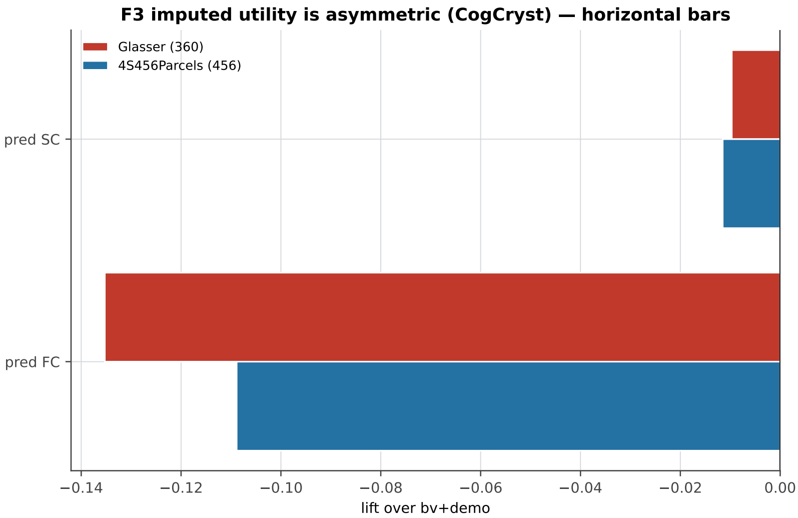
F3 · . . . but their utility is asymmetric

grouped bars



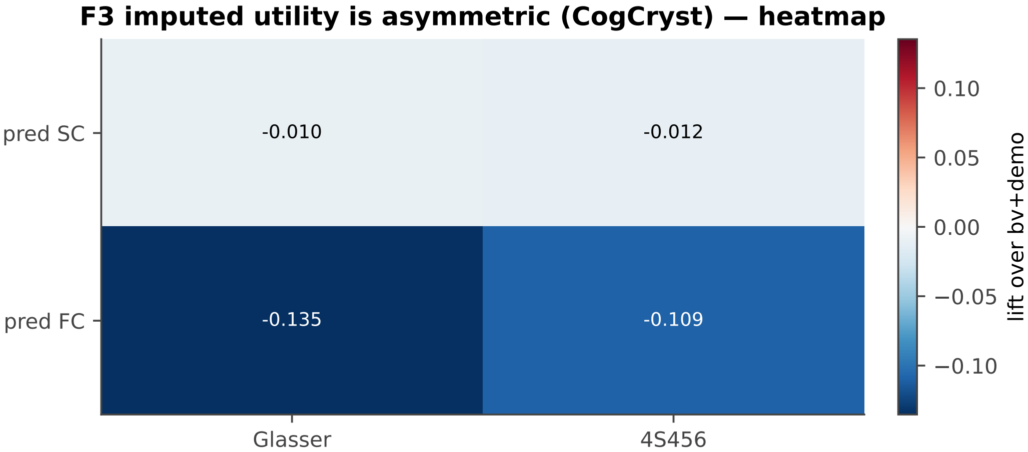
F3 · . . . but their utility is asymmetric

horizontal bars



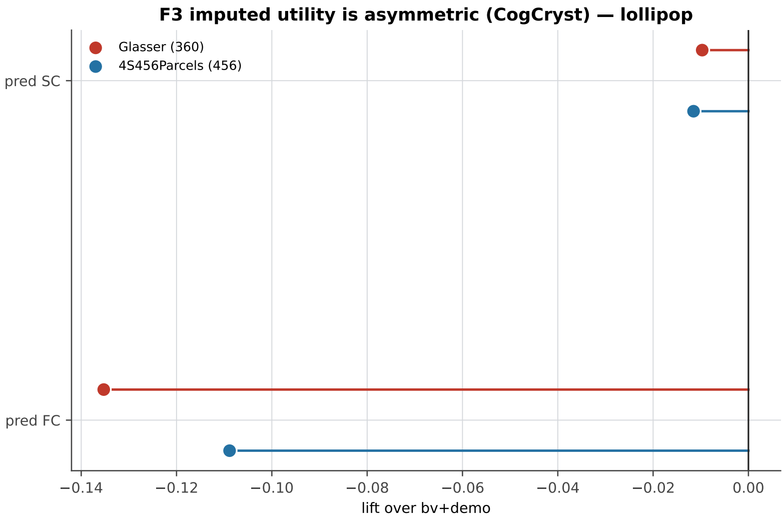
F3 · . . . but their utility is asymmetric

heatmap



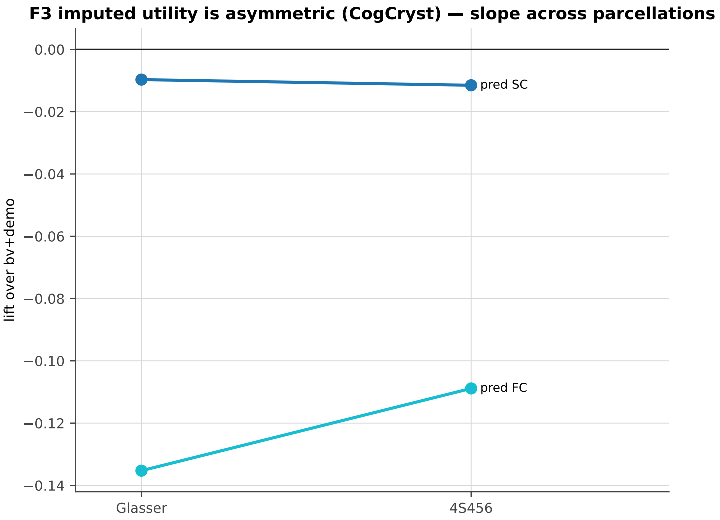
F3 · . . . but their utility is asymmetric

lollipop



F3 · . . . but their utility is asymmetric

slope across parcellations

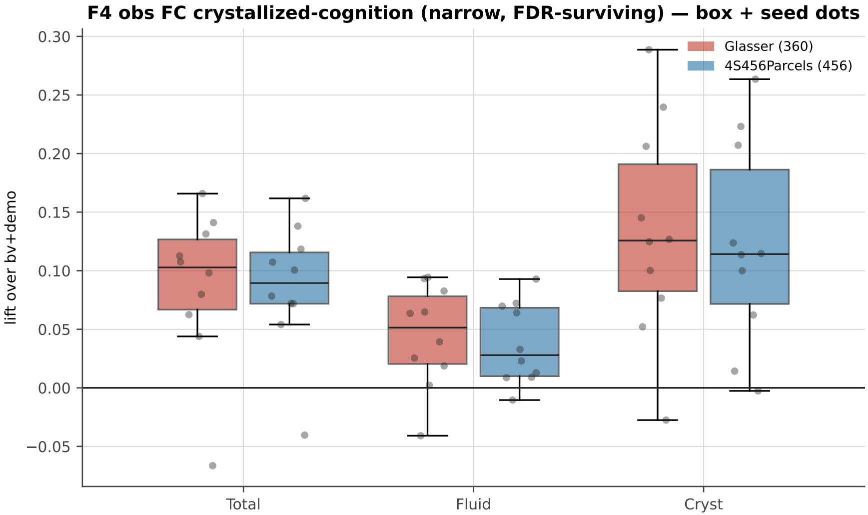


F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

7 chart variants

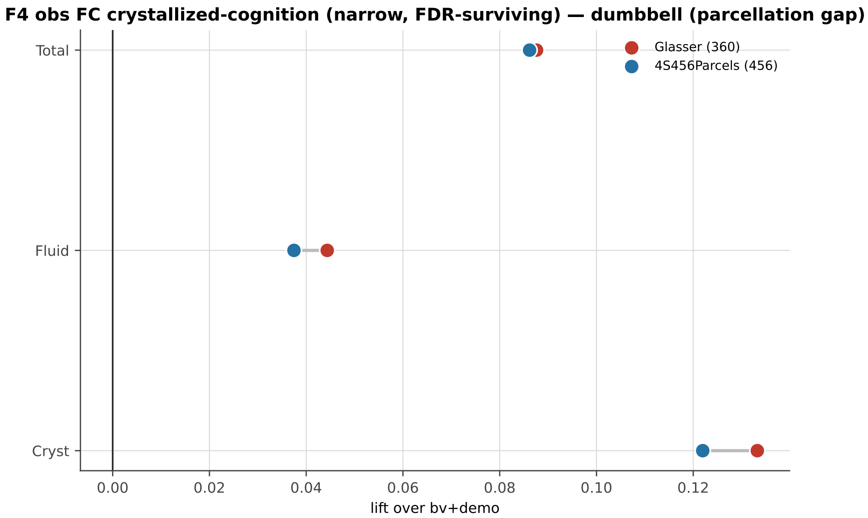
F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

box + seed dots



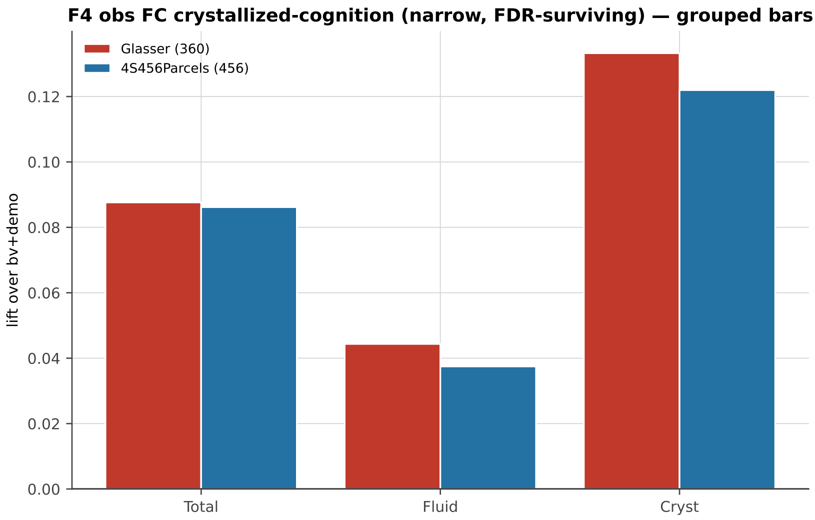
F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

dumbbell (parcellation gap)



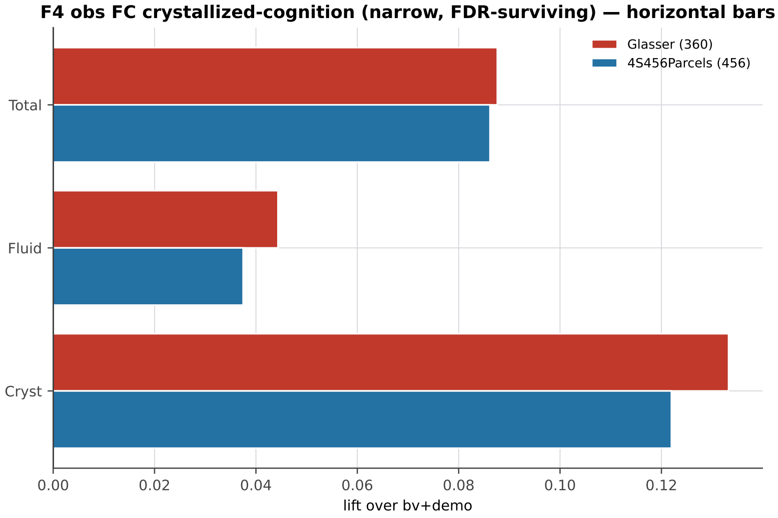
F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

grouped bars



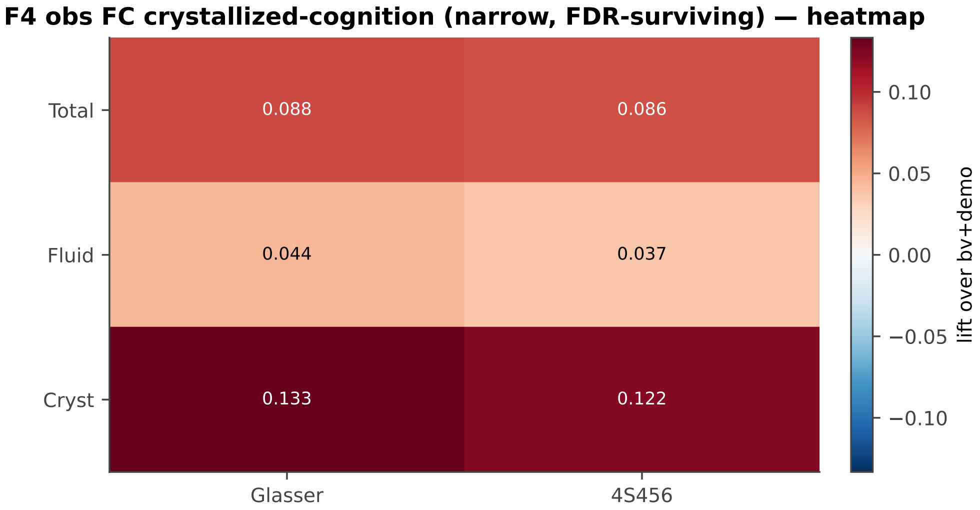
F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

horizontal bars



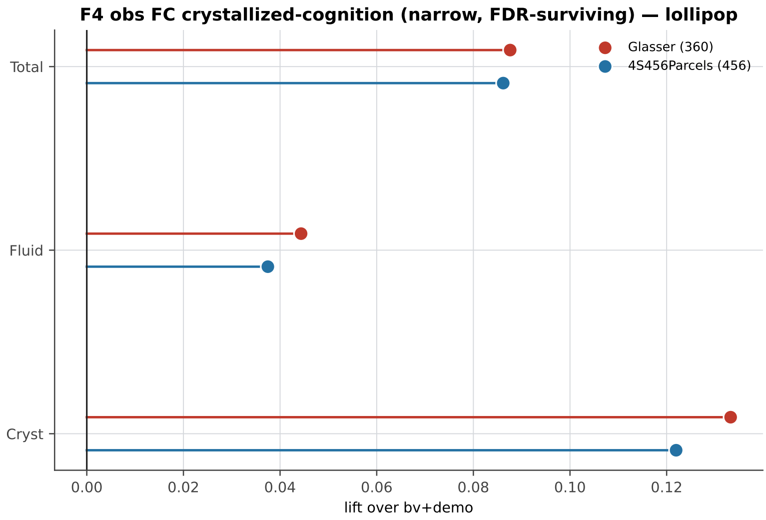
F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

heatmap



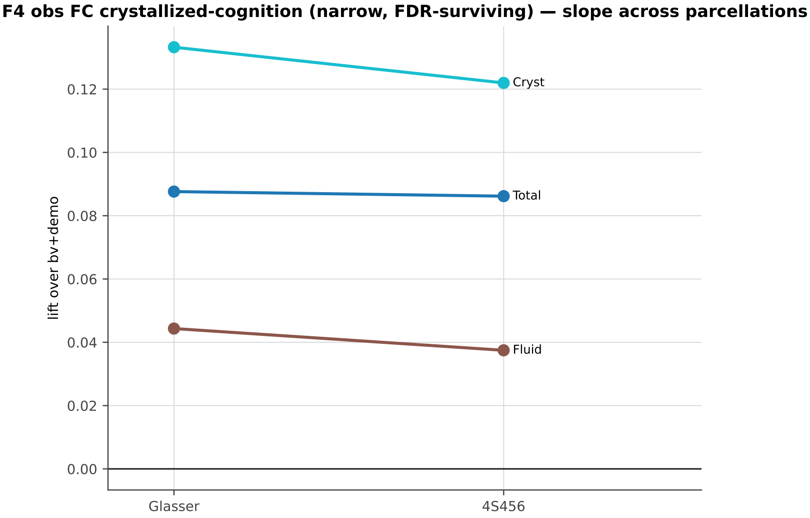
F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

lollipop



F4 · Observed FC: crystallized-cognition signal (narrow, FDR-surviving)

slope across parcellations

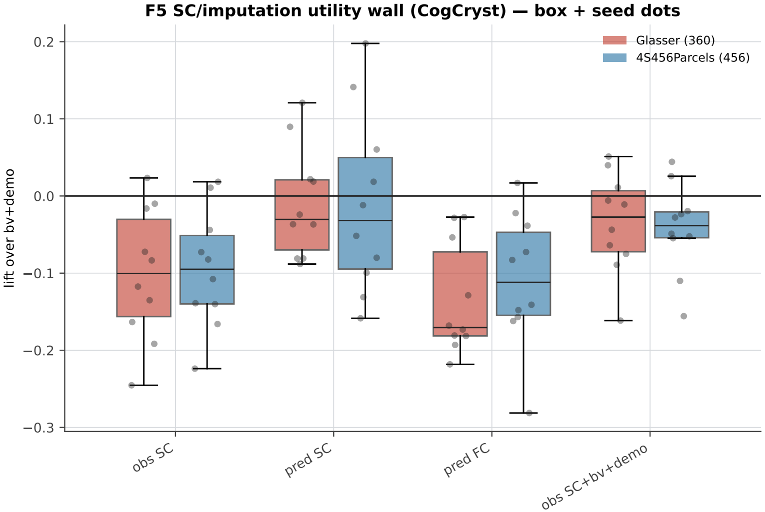


F5 · SC / imputation utility wall

7 chart variants

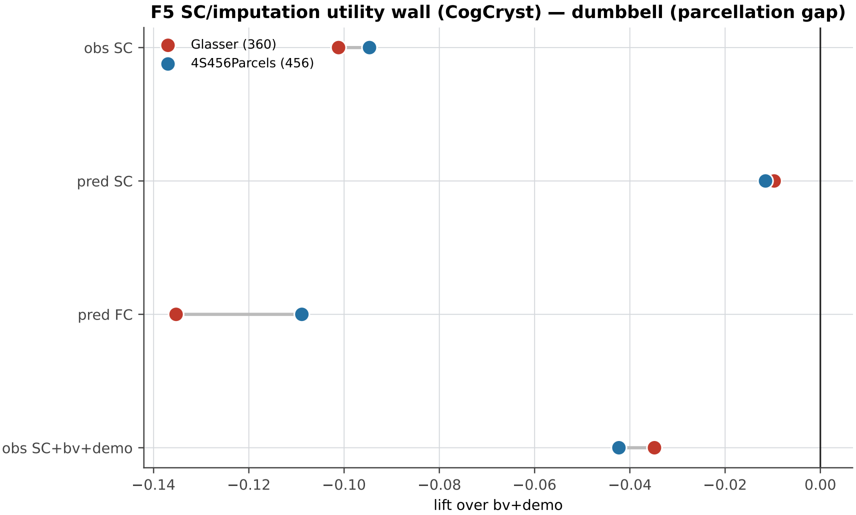
F5 · SC / imputation utility wall

box + seed dots



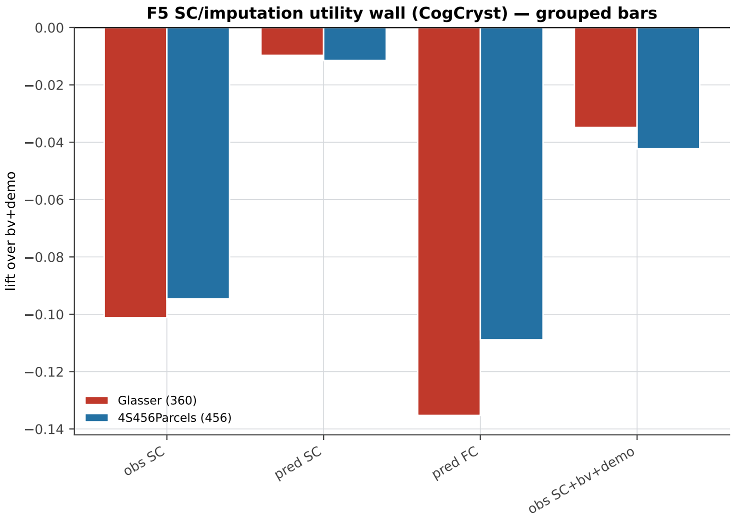
F5 · SC / imputation utility wall

dumbbell (parcellation gap)



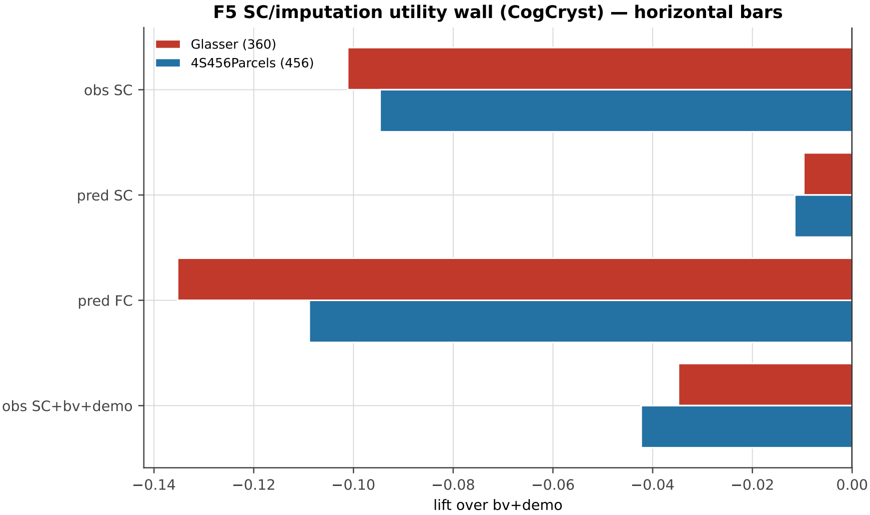
F5 · SC / imputation utility wall

grouped bars



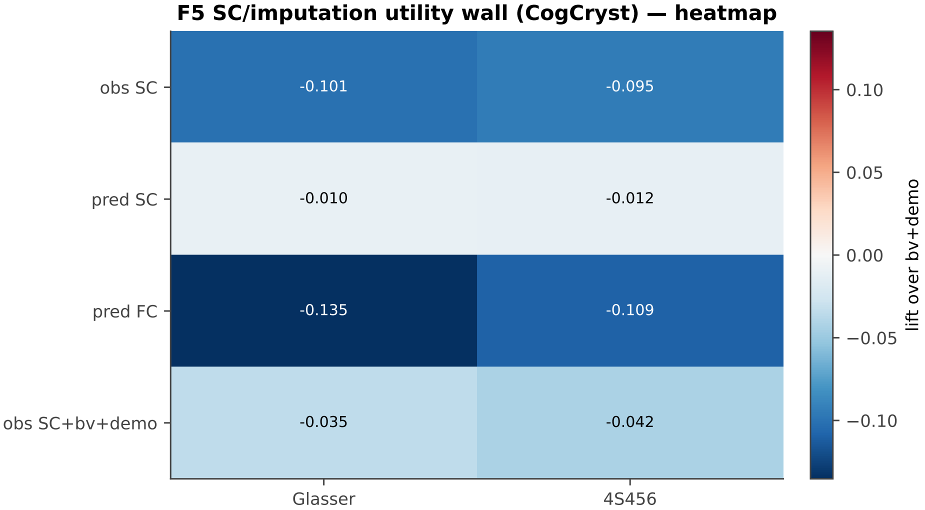
F5 · SC / imputation utility wall

horizontal bars



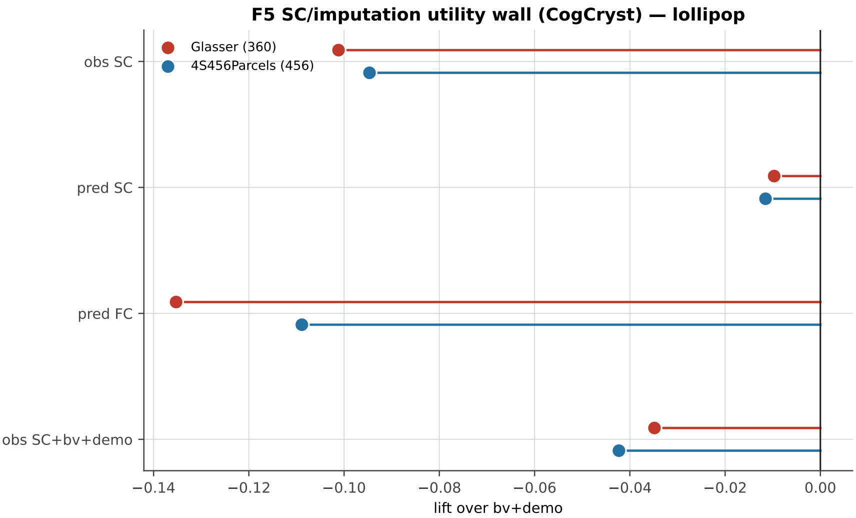
F5 · SC / imputation utility wall

heatmap



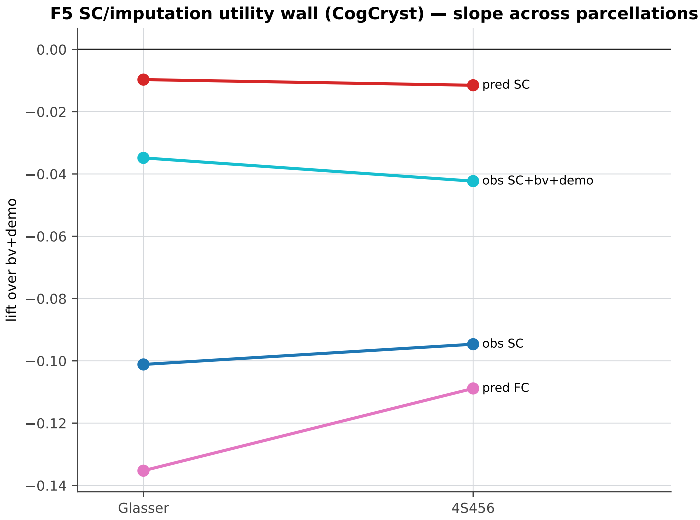
F5 · SC / imputation utility wall

lollipop



F5 · SC / imputation utility wall

slope across parcellations

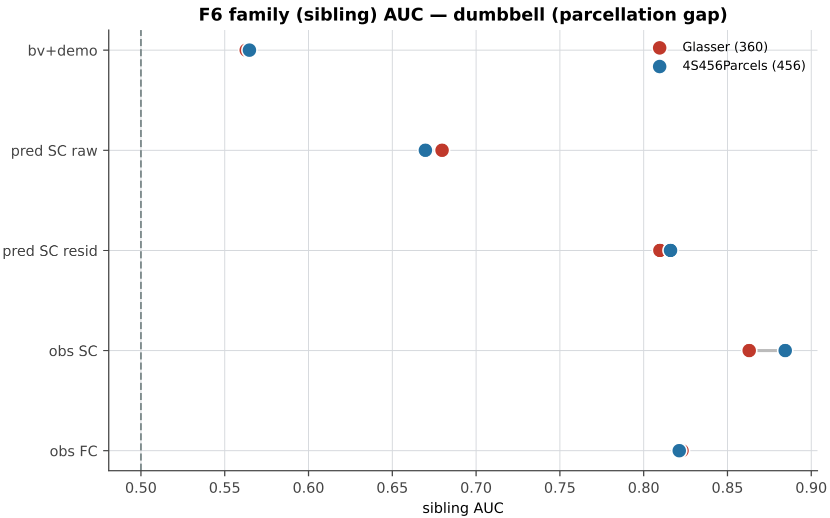


F6 · Predicted connectomes carry family signal

6 chart variants

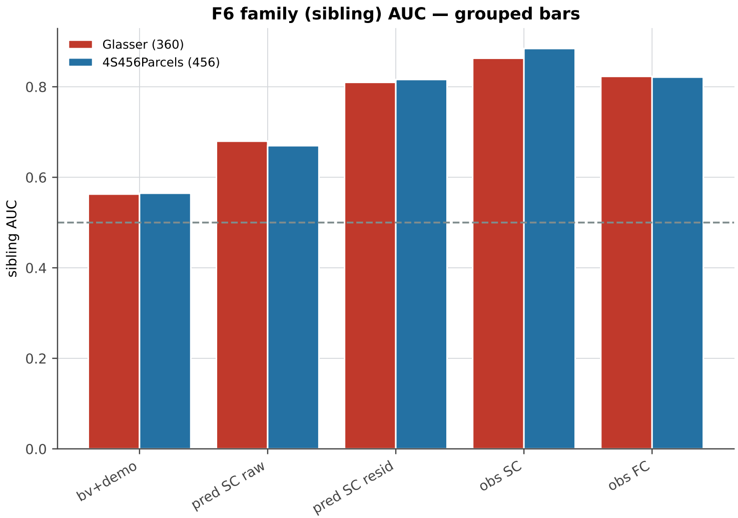
F6 · Predicted connectomes carry family signal

dumbbell (parcellation gap)



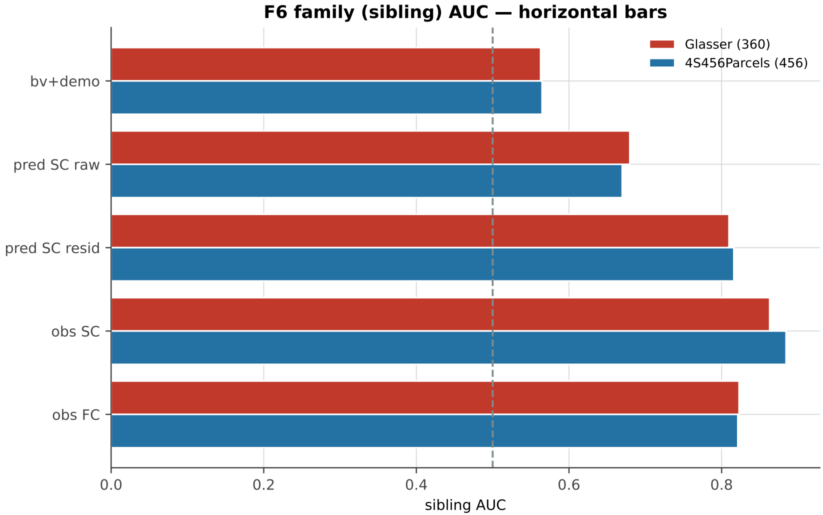
F6 · Predicted connectomes carry family signal

grouped bars



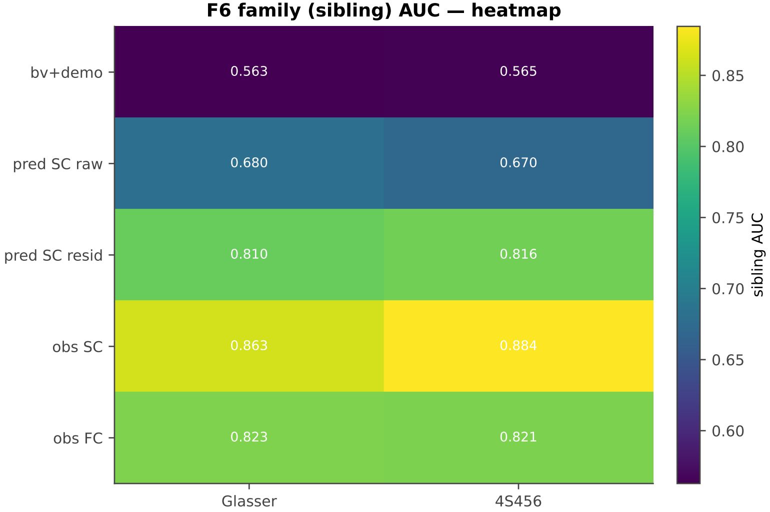
F6 · Predicted connectomes carry family signal

horizontal bars



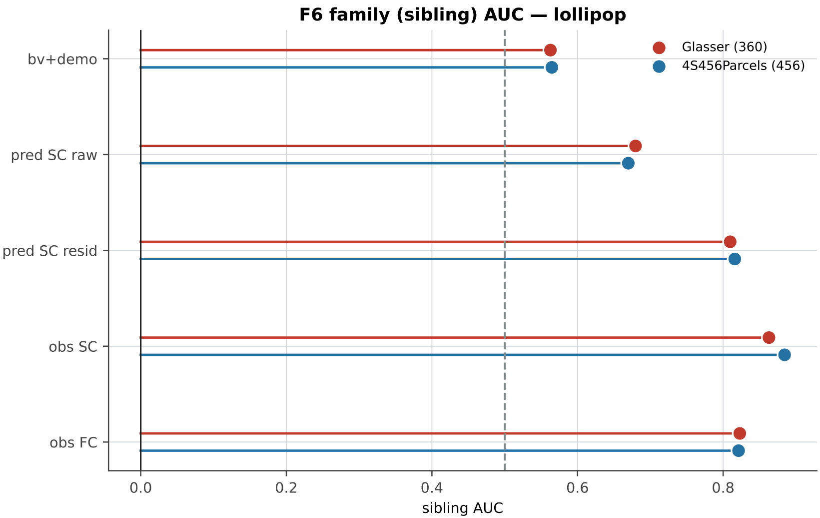
F6 · Predicted connectomes carry family signal

heatmap



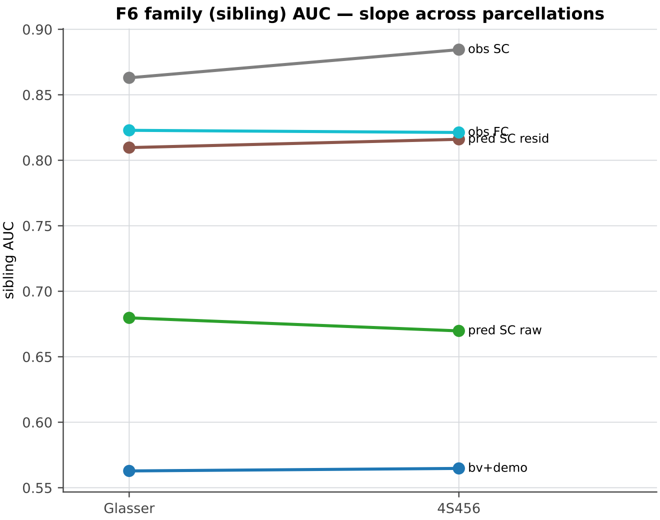
F6 · Predicted connectomes carry family signal

lollipop



F6 · Predicted connectomes carry family signal

slope across parcellations

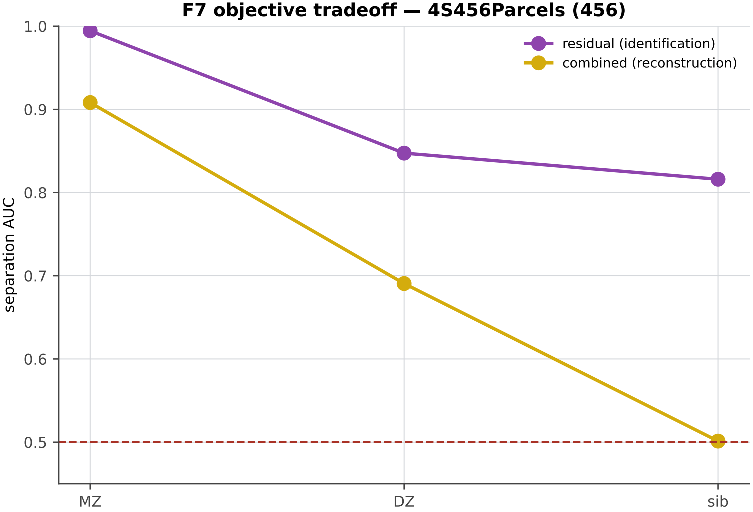


F7 · Reconstruction vs identification tradeoff

2 chart variants

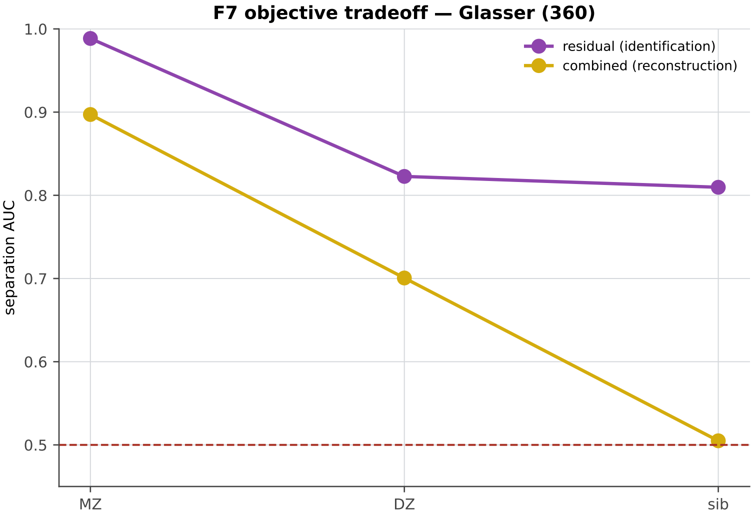
F7 · Reconstruction vs identification tradeoff

4S456Parcels



F7 · Reconstruction vs identification tradeoff

Glasser

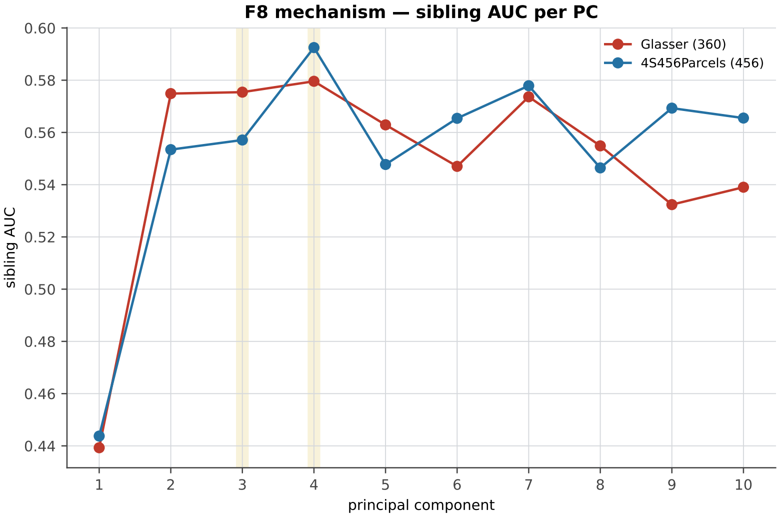


F8 · FC-predictable low-variance SC mode (component index not stable)

3 chart variants

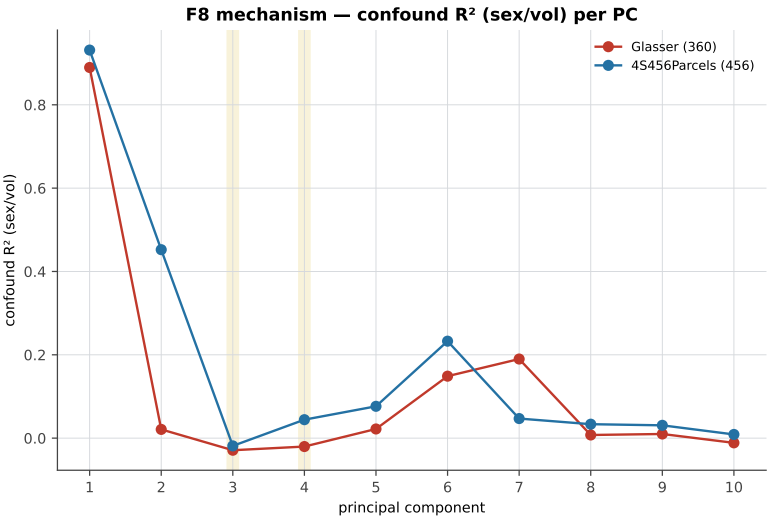
F8 · FC-predictable low-variance SC mode (component index not stable)

sibling AUC per PC



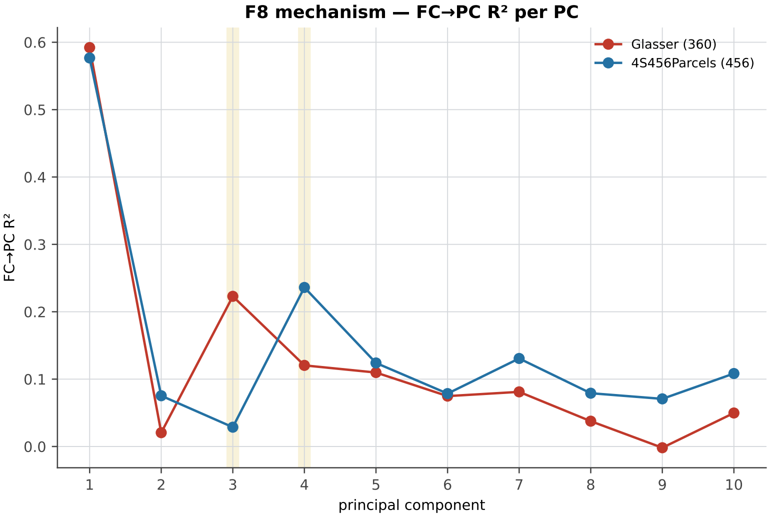
F8 · FC-predictable low-variance SC mode (component index not stable)

sex/volume confound per PC



F8 · FC-predictable low-variance SC mode (component index not stable)

FC→PC predictability per PC

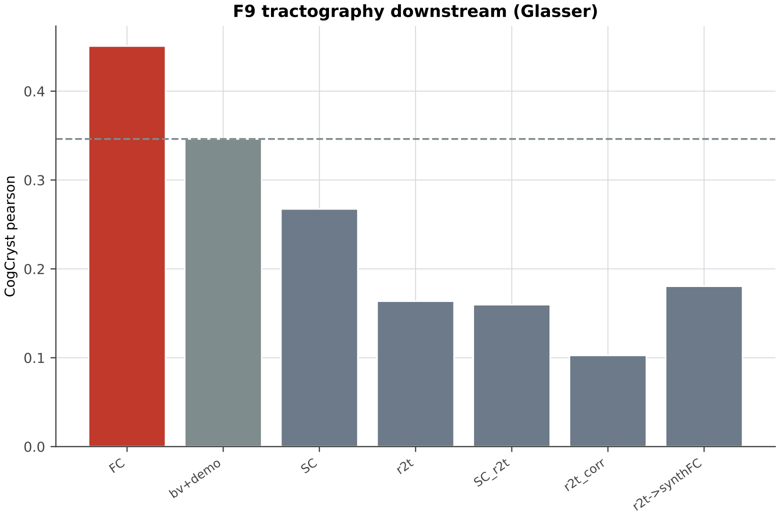


F9 · Richer tractography does not help

3 chart variants

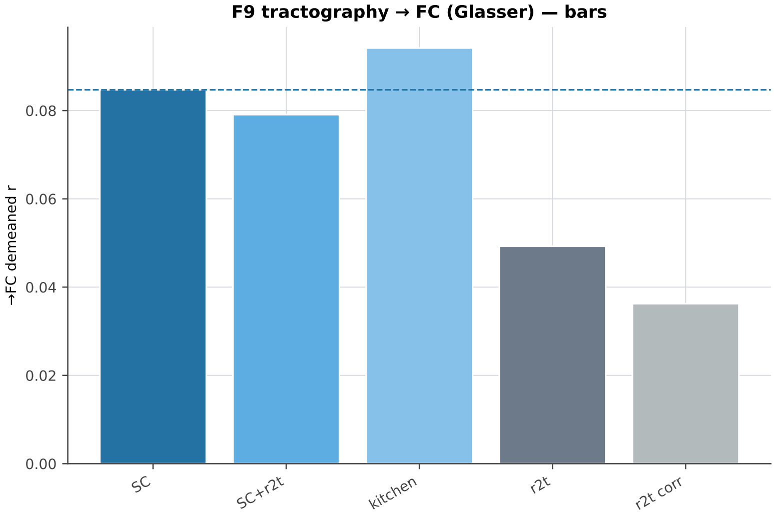
F9 · Richer tractography does not help

downstream cognition



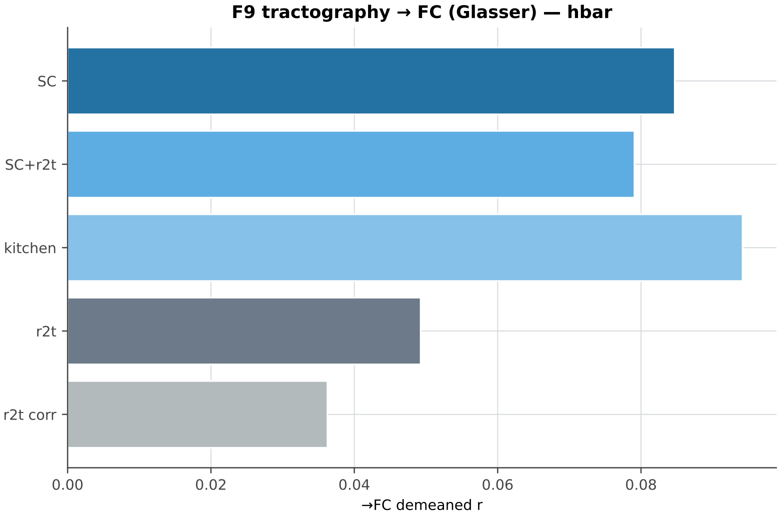
F9 · Richer tractography does not help

→FC bars



F9 · Richer tractography does not help

→FC hbar

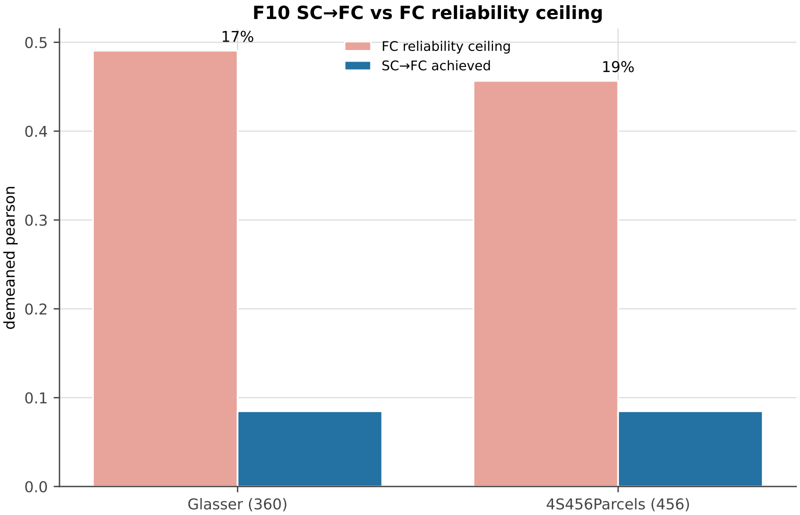


F10 · The ceiling is structural

3 chart variants

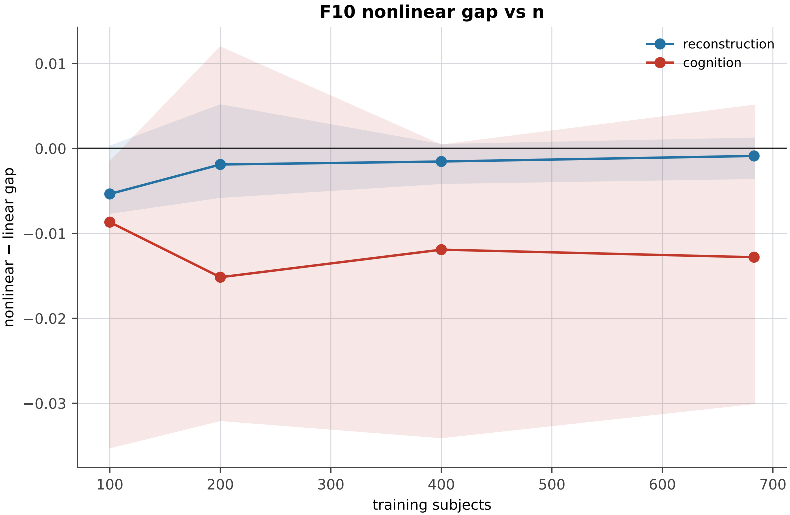
F10 · The ceiling is structural

vs FC reliability ceiling



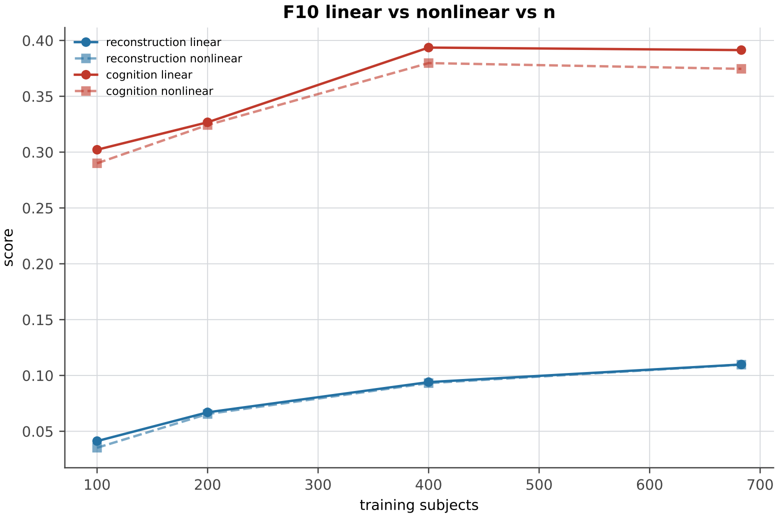
F10 · The ceiling is structural

nonlinear gap vs n



F10 · The ceiling is structural

linear vs nonlinear vs n



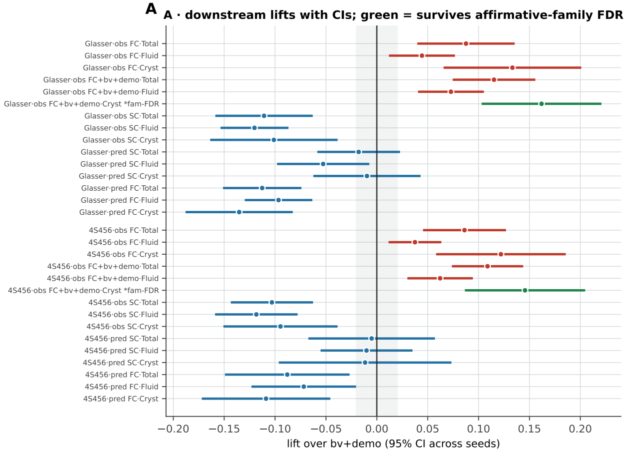
F11 · The cognition lift is multiplicity-fragile

1 chart variants

F11 · The cognition lift is multiplicity-fragile

full composite panel

F11 · The observed-FC cognition lift is real but multiplicity-fragile (my robustness analysis)



B

Multiplicity & power

Median MDE (80% power, n=10 seeds)

0.056 pearson

Positive lifts surviving whole-grid FDR

0 / 54

Positive lifts surviving family FDR

2 / 12

obs FC CogCryst — uncorrected p

≈0.03–0.04

obs FC CogCryst — whole-grid q

≈0.24 (fails)

obs FC+bv+demo CogCryst — family q

0.027/0.048 (passes)

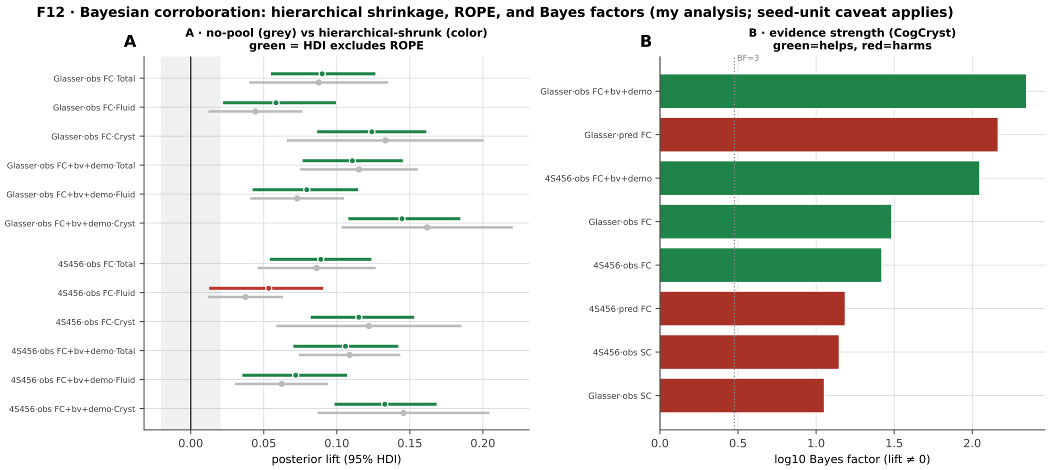
Design well-powered for ~0.1 lifts, underpowered below ~0.03.

F12 · Bayesian corroboration

1 chart variants

F12 · Bayesian corroboration

full composite panel

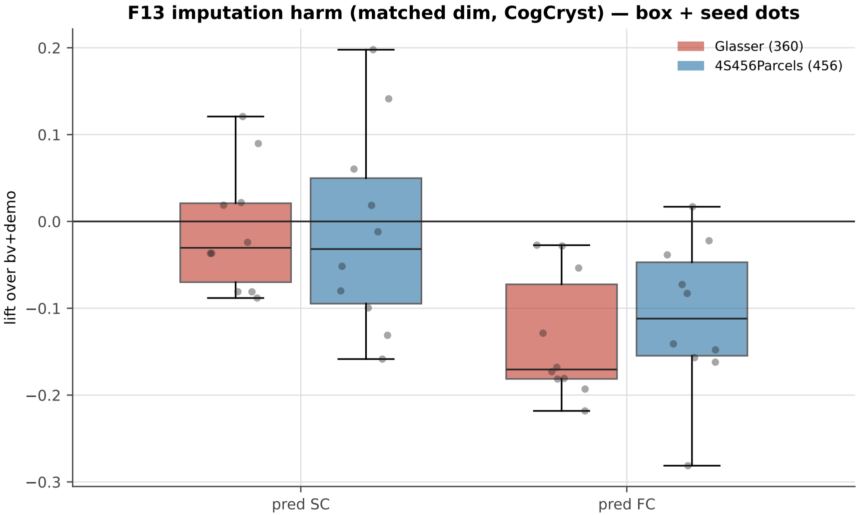


F13 · Imputation harm / cross-modal loss

7 chart variants

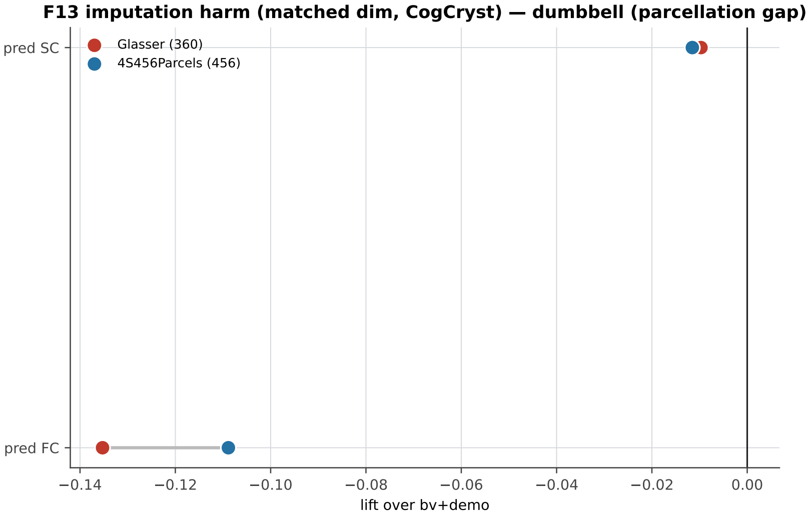
F13 · Imputation harm / cross-modal loss

box + seed dots



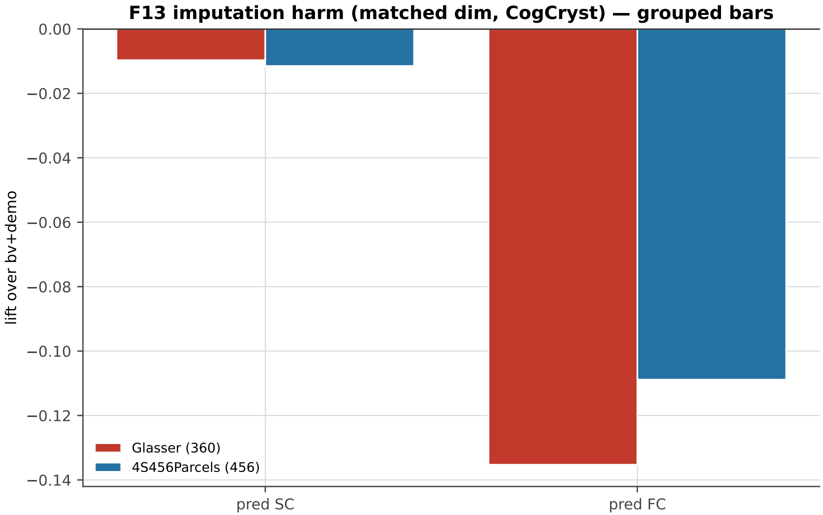
F13 · Imputation harm / cross-modal loss

dumbbell (parcellation gap)



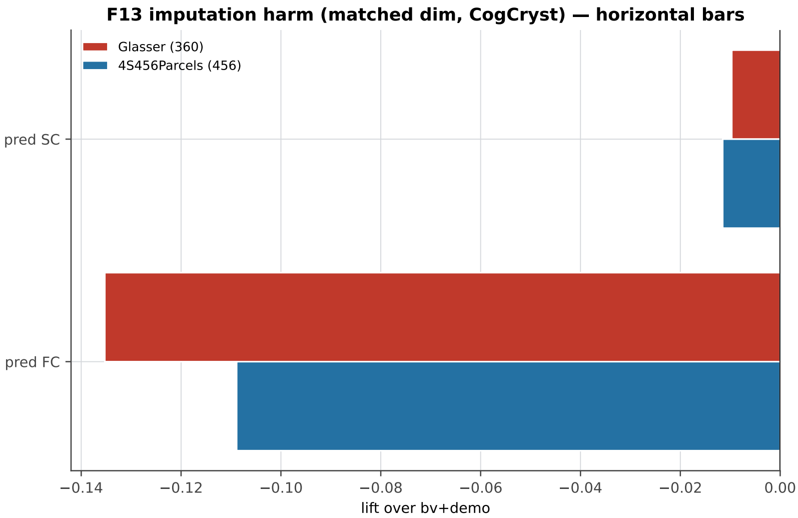
F13 · Imputation harm / cross-modal loss

grouped bars



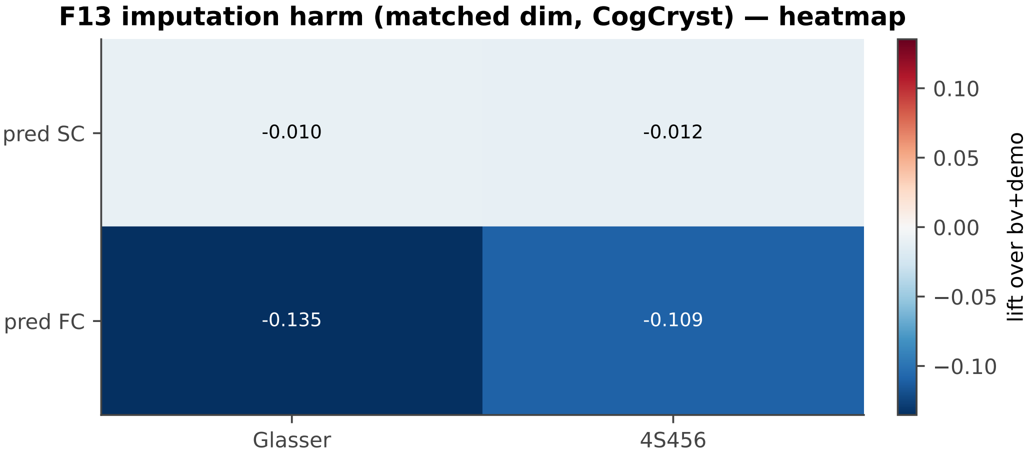
F13 · Imputation harm / cross-modal loss

horizontal bars



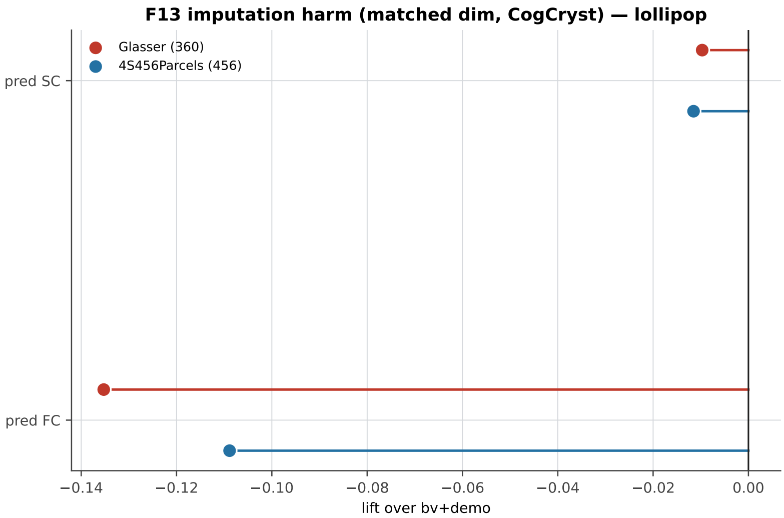
F13 · Imputation harm / cross-modal loss

heatmap



F13 · Imputation harm / cross-modal loss

lollipop



F13 · Imputation harm / cross-modal loss

slope across parcellations